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The Effect of Brensocatib vs Placebo on Symptom Burden in Patients With or Without On-study Pulmonary Exacerbations: A Post hoc Analysis From the ASPEN Trial

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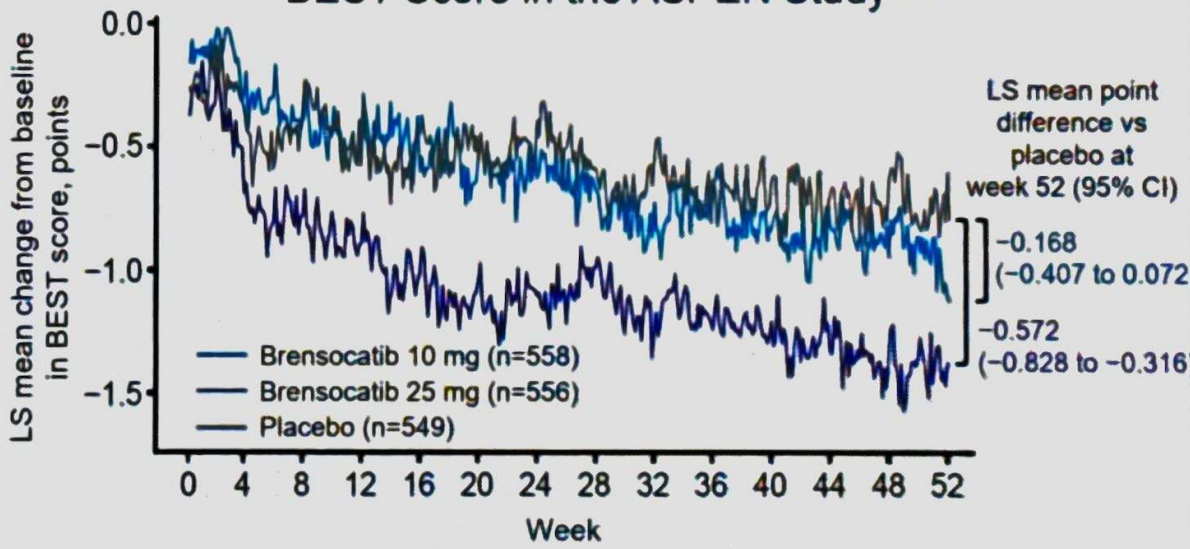
Lesson Objectives

- Report the effect of brensocatib vs placebo on patient-reported symptom burden as measured by the Bronchiectasis Exacerbation and Symptom Tool (BEST) diary in adult patients with vs without on-study exacerbations from the ASPEN trial

ASPEN (NCT04594369): A Phase 3, Randomized, Double-Blind, 52-Week Trial in Patients With Bronchiectasis

- Brensocatib 10 mg and 25 mg **significantly reduced the burden of PEx** over 52 weeks vs placebo in patients with bronchiectasis; brensocatib 25 mg **significantly reduced lung function decline**[1]
- Brensocatib 25 mg also **nominally significantly improved patient-reported symptoms** as measured by the QOL-B RSS and BEST Symptom Diary (exploratory endpoint)[1,2]

LS Mean Change From Baseline in Average Daily BEST Score in the ASPEN Study[a,1,2]



BEST scores range from 0-26, with lower scores indicating fewer symptoms.

[a]Average daily change in the BEST score was analyzed with an analysis of covariance model in adult patients in the ITT analysis. Baseline values are adjusted as a covariate.

1. Chalmers JD, et al. *N Engl J Med*. 2025;392(16):1569-1581. 2. Chalmers JD, et al. 7th World Bronchiectasis Conference; 2024: Abstract 03.07.

BEST, Bronchiectasis Exacerbation and Symptom Tool; ITT, intent-to-treat; LS, least squares; PEx, pulmonary exacerbations; QOL-B RSS, Quality of Life-Bronchiectasis questionnaire Respiratory Symptoms score.

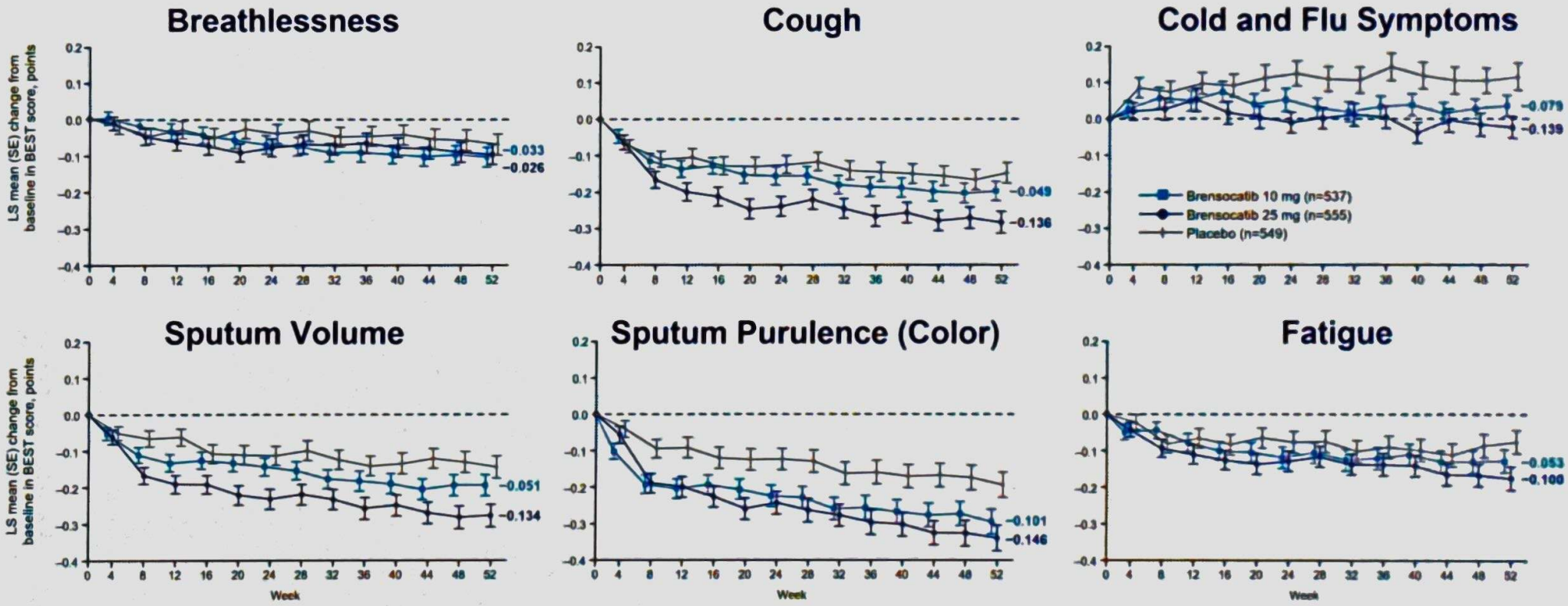
Methods

- Adults (≥18 years) enrolled in ASPEN had ≥2 PEx in the 12 months before screening and were randomized 1:1:1 to once-daily brensocatib (10 mg or 25 mg) or placebo for 52 weeks
- Adult patients completed the BEST diary on a daily basis, which evaluates 6 key symptoms relevant to bronchiectasis and PEx (breathlessness, fatigue, sputum volume, sputum purulence [color], cough, cold/flu symptoms), from screening through end of study
  - Overall scores range from 0-26 points, with lower scores indicating fewer symptoms
- In this post hoc analysis, patients were categorized into 3 subgroups by the number of PEx they had during the study (no PEx, any PEx, any severe PEx)
  - PEx were adjudicated by a blinded independent committee
  - Severe PEx were those requiring IV antibiotics and/or hospitalization

PEx, pulmonary exacerbations.

Change From Baseline in Individual BEST Item Scores in the Overall Adult Population

- Both doses of brensocatib provided greater improvements than placebo across all BEST items

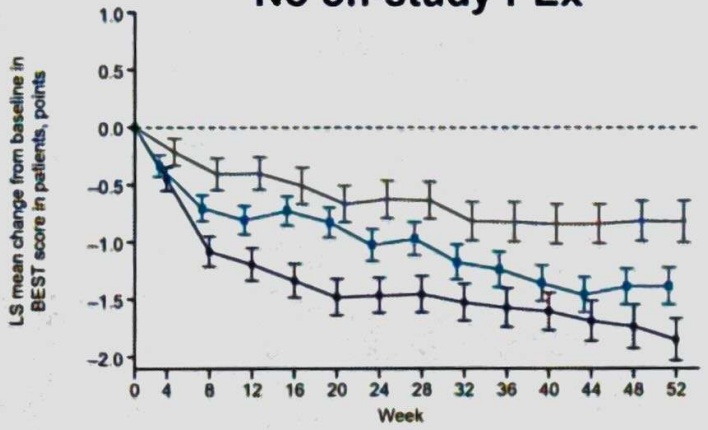


For each BEST item, lower scores indicate fewer symptoms. Data values to the right of each graph are the LS mean point difference vs placebo at week 52.
BEST, Bronchiectasis Exacerbation and Symptom Tool; LS, least squares.

Change From Baseline in Overall BEST Scores by On-Study Exacerbation Status

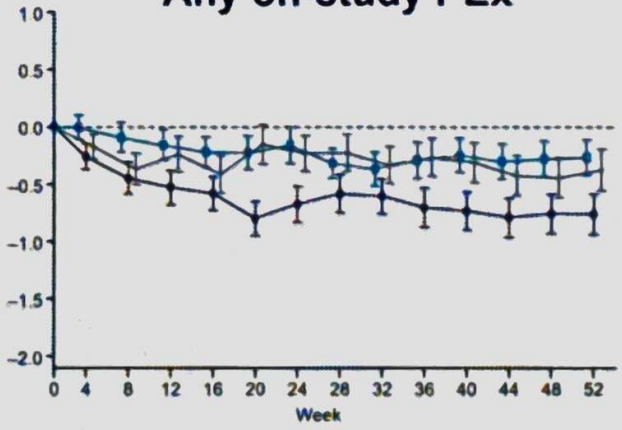
- The greatest symptom burden was observed in patients with PEx and specifically severe PEx
- Brensocatib 25 mg consistently provided greater improvements vs placebo regardless of PEx occurrence or severity

No on-study PEx



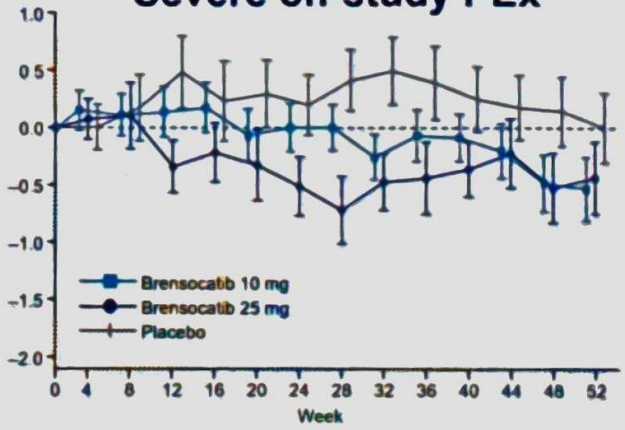
	Brensocatib 10 mg n=278[a]	Brensocatib 25 mg n=276[a]	Placebo n=253[a]
LS mean (SE) change from baseline at week 52	−1.39 (0.16)	−1.86 (0.19)	−0.83 (0.18)
LS mean (95% CI) difference vs placebo at week 52	−0.57 (−1.05 to −0.08)	−1.03 (−1.55 to −0.52)	-

Any on-study PEx



	Brensocatib 10 mg n=283[a]	Brensocatib 25 mg n=286[a]	Placebo n=218[a]
LS mean (SE) change from baseline at week 52	−0.32 (0.15)	−0.77 (0.17)	−0.37 (0.17)
LS mean (95% CI) difference vs placebo at week 52	0.06 (−0.39 to 0.50)	−0.40 (−0.87 to 0.07)	-

Severe on-study PEx

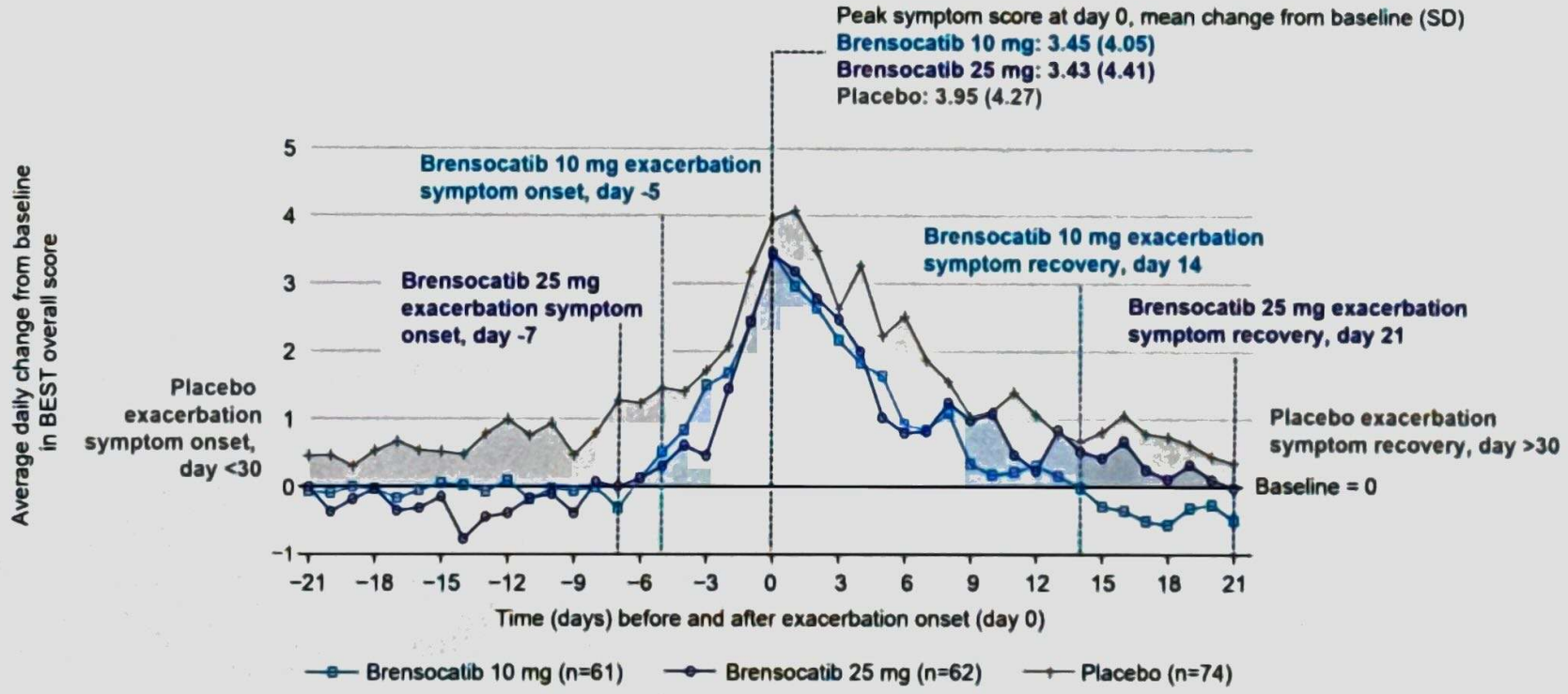


	Brensocatib 10 mg n=[a]	Brensocatib 25 mg n=[a]	Placebo n=74[a]
LS mean (SE) change from baseline at week 52	−0.64 (0.34)	−0.52 (0.38)	0.00 (0.37)
LS mean (95% CI) difference vs placebo at week 52	−0.64 (−1.64 to 0.35)	−0.53 (−1.55 to 0.50)	-

[a]n is the number of patients with observations used in the analysis. Average daily change in the BEST score was analyzed with an analysis of covariance model. BEST scores range from 0-26, with lower scores indicating fewer symptoms. Baseline values are adjusted as a covariate.
BEST, Bronchiectasis Exacerbation and Symptom Tool; LS, least squares; PEx, pulmonary exacerbations.

Average Daily Change From Baseline in Overall BEST Score Around Severe PEx Events

- Both doses of brensocatib reduced symptom burden of severe PEx vs placebo



Day 0 was defined as the start of the exacerbation as determined by an independent adjudication committee. Symptom onset was defined as the time point after a sustained mean increase from study baseline (mean change = 0) in BEST score by >0 points for ≥48 hours before day 0. Symptom recovery was defined as the time point when symptoms returned to baseline (mean change from baseline in BEST score ≤0 points) after day 0.

BEST, Bronchiectasis Exacerbation and Symptom Tool; LS, least squares; PEx, pulmonary exacerbations.

Conclusions

- Brensocatib reduced symptom burden, regardless of exacerbation occurrence or severity and across all BEST items
  - Improvements were greatest in patients treated with brensocatib 25 mg and patients without on-study exacerbation
- Brensocatib 10 mg and 25 mg also reduced the symptom burden vs placebo during severe exacerbation events
- Patient-reported symptom burden was lowest in patients without exacerbations, highlighting the importance of exacerbation prevention

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BEST, Bronchiectasis Exacerbation and Symptom Tool.